

A Theoretical Derivation about the impact of redundant information

To characterize how redundant information disrupts LLM reasoning, we quantify this via symbolic derivation.

We consider two distinct scenarios. In the first, LVLs perform multimodal reasoning and analysis within a context devoid of interference, where both the input images and textual content are solely intended to provide evidence for problem-solving. In contrast, real-world scenarios are considerably more complex: beyond useful cues, the input often contains a substantial amount of irrelevant content. At the visual level, this manifests as numerous unrelated images, where relevant images are interspersed among all images and jointly provided as input to the LLM.

Let M denote the number of relevant tokens received by the model in the idealized setting, where the input contains only the question string and the visually relevant pages associated with the question, collectively covering all evidence required for question answering; this corresponds to a clean-case scenario. In the presence of N irrelevant visual tokens, the total number of tokens becomes $M + N$. This redundancy serves to divert the model's attention from the relevant tokens to the irrelevant ones, thereby contaminating the hidden representations and diminishing the effectiveness of the information.

However, it is important to emphasize that such redundant information must exhibit a certain degree of confusability. Specifically, the visual features of the irrelevant images should bear strong similarity to the target problem, thereby inducing a high level of distraction and semantic ambiguity. Only under this condition do the redundant tokens possess substantial semantic attraction; otherwise, the model will allocate negligible attention to these tokens.

We start from **the first layer**. Let the attention computation for the x -th token in the clean case be represented as h_x^1 . Upon the introduction of N irrelevant visual tokens, the hidden state for the x -th token after the attention layer evolves to $h_x'^1$:

$$h_x^1 = \hat{h}_x^0 + \alpha_1 V_1 + \alpha_2 V_2 + \dots + \alpha_M V_M, \quad (12)$$

$$h_x'^1 = \hat{h}_x^0 + \beta_1 V'_1 + \beta_2 V'_2 + \dots + \beta_{M+N} V'_{M+N}, \quad (13)$$

where $\hat{h}_x^0$ and $\hat{h}_x'^0$ denote the output states of the previous decoder block (here, the initial embeddings), representing the effect of the residual connection, α_i, β_i represent the attention weights, and $V_i, V'_i \in \mathbb{R}^d$ are the *value* vectors at the i -th position. The order of the tokens does not affect the derivation; for simplicity, we assume that the first M tokens are relevant, while the subsequent N tokens are irrelevant.

Due to the normalization of attention scores, both $\sum_{i=1}^M \alpha_i$ and $\sum_{j=1}^{M+N} \beta_j$ are equal to 1. After the addition of redundant tokens, the attention scores for the M relevant tokens are diluted and redistributed across the N irrelevant tokens. To standardize the notation, let the transformation of the average scores before and after the redistribution in relevant tokens be expressed as:

$$\beta_i = \frac{M}{M+N} \cdot \delta \alpha_i, \quad i = 1, 2, \dots, M, \quad (14)$$

where δ is a harmonic coefficient. For convenience, we assume that the δ value is equal across all tokens after averaging. Given that the model inherently exhibits a strong bias towards relevant

information, the degree of attenuation of the attention scores for the M relevant tokens is not as severe as $\frac{M}{M+N}$ compared to the clean case, hence $\delta > 1$.

Therefore, the expression for h_x' in the first layer of the decoder block is given by:

$$\begin{aligned} h_x'^1 &= \hat{h}_x^0 + \beta_1 V'_1 + \beta_2 V'_2 + \dots + \beta_{M+N} V'_{M+N} \\ &= \hat{h}_x^0 + \frac{\delta M}{M+N} (\alpha_1 V'_1 + \alpha_2 V'_2 + \dots + \alpha_M V'_M) \\ &\quad + \beta_{M+1} V'_{M+1} + \dots + \beta_{M+N} V'_{M+N} \\ &= \hat{h}_x^0 + \frac{\delta M}{M+N} \sum_{i=1}^M \alpha_i V'_i + \sum_{j=M+1}^{M+N} \beta_j V'_j, \quad \delta > 1. \end{aligned} \quad (15)$$

The computation of h_x' above reflects the fact that, in the presence of redundant context, after the first layer of attention, the hidden state at each token position will be mixed with information derived from the irrelevant tokens. As a result, the information purity of the hidden state is diminished.

We quantify the **information purity** of the x -th token after the first layer as η , which represents the proportion of information from relevant tokens in the x -th hidden state after the first layer of attention, relative to the total value of the x -th hidden state:

$$\begin{aligned} \eta &= \frac{\hat{h}_x^0 + \frac{\delta M}{M+N} \sum_{i=1}^M \alpha_i V'_i}{\hat{h}_x^0 + \frac{\delta M}{M+N} \sum_{i=1}^M \alpha_i V'_i + \sum_{j=M+1}^{M+N} \beta_j V'_j} \\ &= \frac{1 + \frac{\hat{h}_x^0}{\frac{\delta M}{M+N} \sum_{i=1}^M \alpha_i V'_i}}{1 + \frac{\hat{h}_x^0}{\frac{\delta M}{M+N} \sum_{i=1}^M \alpha_i V'_i} + \frac{M+N}{\delta M} \cdot \frac{\sum_{j=M+1}^{M+N} \beta_j V'_j}{\sum_{i=1}^M \alpha_i V'_i}} \\ &= \frac{1 + \frac{\hat{h}_x^0}{\sum_{i=1}^M \beta_i V'_i}}{1 + \frac{\hat{h}_x^0}{\sum_{i=1}^M \beta_i V'_i} + \frac{M+N}{\delta M} \cdot \frac{\sum_{j=M+1}^{M+N} \beta_j V'_j}{\sum_{i=1}^M \alpha_i V'_i}} \\ &\in (0, 1), \quad \delta > 1. \end{aligned} \quad (16)$$

At this point, we have obtained a dilution factor $\eta \in (0, 1)$. Since $\frac{\sum_{j=M+1}^{M+N} \beta_j V'_j}{\sum_{i=1}^M \alpha_i V'_i}$ represents the weighted value of the N redundant tokens divided by the weighted value of the M relevant tokens, it is primarily influenced by the ratio of M to N . In the extreme case, this ratio approaches $\frac{N}{M}$. However, due to the model's inherent resistance to redundant information, the attention weights β allocated to the redundant tokens will be lower than those of the relevant tokens. As a result, the numerator will be smaller than the ideal case. Moreover, the weights in the denominator correspond to the original attention weights α of the M relevant tokens, rather than the diluted weights β , which further increases the denominator. Consequently, the ratio cannot fully reach the value of $\frac{N}{M}$. To account for this, we introduce a harmonic coefficient λ , yielding a ratio of $\frac{\lambda N}{M}$, where $0 < \lambda < 1$.

Therefore, the original equation (16) can be expressed as:

$$\begin{aligned}\eta &= \frac{1 + \frac{\hat{h}_x^{r0}}{\sum_{i=1}^M \beta_i V_i'}}{1 + \frac{\hat{h}_x^{r0}}{\sum_{i=1}^M \beta_i V_i'} + \frac{M+N}{\delta M} \cdot \frac{\sum_{j=M+1}^{M+N} \beta_j V_j'}{\sum_{i=1}^M \alpha_i V_i'}} \\ &\Rightarrow \frac{1 + \frac{\hat{h}_x^{r0}}{\sum_{i=1}^M \beta_i V_i'}}{1 + \frac{\hat{h}_x^{r0}}{\sum_{i=1}^M \beta_i V_i'} + \frac{M+N}{\delta M} \cdot \frac{\lambda N}{M}} \\ &\in (0, 1), \quad \delta > 1, \lambda \in (0, 1).\end{aligned}\quad (17)$$

By analogy, referring to Eqs. (16), on the x -th token, after reaching the second layer, the ratio of the value originating from relevant information in the hidden state to the total value is given by:

$$\eta_2 = \frac{(\hat{h}_x^{r1} + \frac{\delta M}{M+N} \sum_{i=1}^M \alpha_i^2 V_i'^2) \cdot \eta}{\hat{h}_x^{r1} + \frac{\delta M}{M+N} \sum_{i=1}^M \alpha_i^2 V_i'^2 + \sum_{j=M+1}^{M+N} \beta_j^2 V_j'^2}.\quad (18)$$

In the formula, “2” denotes “the second layer”, not “square”. Here, we explain why each value in the numerator must be multiplied by the dilution factor η from the first layer: At this point, even the hidden states at the positions of the relevant tokens are contaminated by redundant information from the previous layer. Not all the information at these positions is relevant, and the proportion of relevant information has already been computed as η . Therefore, the actual relevant information at the i -th position in this layer is $\hat{h}_i^{r1} W_V^2 \cdot \eta = V_i'^2 \cdot \eta$, which is why we multiply each value in the numerator by η .

Finally, analogous to Eqs. (16) and (17) in the first layer, simplifying Eq. (18) yields the information purity after the second layer:

$$\begin{aligned}\eta_2 &= \frac{(\hat{h}_x^{r1} + \frac{\delta M}{M+N} \sum_{i=1}^M \alpha_i^2 V_i'^2) \cdot \eta}{\hat{h}_x^{r1} + \frac{\delta M}{M+N} \sum_{i=1}^M \alpha_i^2 V_i'^2 + \sum_{j=M+1}^{M+N} \beta_j^2 V_j'^2} \\ &\Rightarrow \frac{1 + \frac{\hat{h}_x^{r1}}{\sum_{i=1}^M \beta_i^2 V_i'^2}}{1 + \frac{\hat{h}_x^{r1}}{\sum_{i=1}^M \beta_i^2 V_i'^2} + \frac{M+N}{\delta M} \cdot \frac{\lambda N}{M}} \cdot \eta \\ &= \frac{1 + \frac{\hat{h}_x^{r0} \cdot \eta}{\sum_{i=1}^M \beta_i V_i' \cdot \eta}}{1 + \frac{\hat{h}_x^{r0} \cdot \eta}{\sum_{i=1}^M \beta_i V_i' \cdot \eta} + \frac{M+N}{\delta M} \cdot \frac{\lambda N}{M}} \cdot \eta \\ &= \frac{1 + \frac{\hat{h}_x^{r0}}{\sum_{i=1}^M \beta_i V_i'}}{1 + \frac{\hat{h}_x^{r0}}{\sum_{i=1}^M \beta_i V_i'} + \frac{M+N}{\delta M} \cdot \frac{\lambda N}{M}} \cdot \eta = \eta^2 \in (0, 1).\end{aligned}\quad (19)$$

In other words, after each decoder block, the proportion of relevant information in the hidden state (the information purity) is scaled by η . Consequently, for an L -layer decoder, the effective information retained in the final hidden state is only η^L of its original value, leading to an exponential compression of the informative content:

$$\eta_L = \eta^L, \quad L \in \mathbb{Z}.\quad (20)$$

Furthermore, we analyze the influence of M and N on η :

From the result in Eq. (17), we can observe that the term $1 + \frac{\hat{h}_x^{r0}}{\sum_{i=1}^M \beta_i V_i'}$ appears entirely in the denominator and constitutes a constant greater than 1. For notational simplicity, we denote this

constant by b . We focus on the remaining denominator term $\frac{M+N}{\delta M} \cdot \frac{\lambda N}{M}$:

$$\begin{aligned}\eta &\Rightarrow \frac{1 + \frac{\hat{h}_x^{r0}}{\sum_{i=1}^M \beta_i V_i'}}{1 + \frac{\hat{h}_x^{r0}}{\sum_{i=1}^M \beta_i V_i'} + \frac{M+N}{\delta M} \cdot \frac{\lambda N}{M}} \\ &= \frac{b}{b + \frac{M+N}{\delta M} \cdot \frac{\lambda N}{M}} = \frac{b}{b + \frac{\lambda MN + \lambda N^2}{\delta M^2}} = \frac{b}{b + \frac{\lambda \frac{N}{M} + \lambda (\frac{N}{M})^2}{\delta}} \\ &= \frac{b}{b + \frac{\lambda}{\delta} \left(\frac{N}{M} + (\frac{N}{M})^2 \right)}.\end{aligned}\quad (21)$$

As shown above, the denominator reduces to a univariate quadratic form $x^2 + x$, where $x = \frac{N}{M}$. Since $\frac{N}{M} > 0$ and the function $x^2 + x$ is monotonically increasing for $x > 0$, the denominator increases monotonically as $\frac{N}{M}$ grows, i.e., as redundant tokens constitute a larger proportion of the input. Consequently, η decreases monotonically, indicating a degradation in information purity.

We therefore conclude that as the ratio of redundant tokens to relevant tokens increases, the purity of information in the hidden states decreases, which aligns with intuitive expectations.

Even if η is a value close to 1, its impact under exponentiation can still be substantial. Given the increasing number of layers in modern LLMs and vision encoders, this phenomenon warrants more attention. The contamination of the hidden state with redundant information as additional tokens are incorporated hampers the ability to capture useful signals, ultimately disrupting both vocabulary decoding and output generation.

Given the inevitability of redundancy in real-world data, we argue that an LVLM's capacity to maintain reasoning performance under redundant conditions—rather than under an unrealistically clean setting—should be given greater emphasis as an extended evaluation criterion. Our benchmark is designed precisely for this setting.